Assignment Summary: Data Exploration and Cleaning Using Pandas

This assignment focused on learning the fundamentals of **Python's Pandas library** by performing data exploration and preprocessing on a CSV dataset. Pandas is one of the most widely used libraries in data analysis because it provides powerful tools for handling, cleaning, and manipulating structured data. The objective was to understand how to load a dataset, inspect its contents, identify and resolve data quality issues, perform basic data manipulation, and save the processed dataset for future use.

The first step involved importing the Pandas library and loading the CSV dataset into a **DataFrame**, which is Pandas' primary data structure for storing tabular data. Loading the dataset into a DataFrame made it possible to efficiently access, analyze, and modify the data using built-in Pandas functions.

After successfully loading the dataset, an initial exploration of the data was carried out. Functions such as head() and tail() were used to display the first and last few records, providing a quick overview of the dataset. The shape attribute was used to determine the total number of rows and columns, while the columns attribute displayed all column names. The info() and dtypes functions were used to understand the data types of each column and identify whether any missing values were present. This exploratory analysis helped in understanding the overall structure and characteristics of the dataset before performing any modifications.

The next stage involved identifying and handling missing values. Missing data can reduce the quality of analysis and may affect the accuracy of machine learning models. The isnull().sum() function was used to count the missing values in each column. Depending on the nature of the missing data, appropriate preprocessing techniques such as filling missing values with suitable defaults or removing incomplete records using fillna() or dropna() were applied. This step ensured that the dataset became more complete and suitable for further analysis.

Basic data manipulation operations were then performed using Pandas. Specific columns were selected to focus on important information, while filtering operations were applied to retrieve records that satisfied certain conditions, such as selecting rows with sales above a specific value or filtering products belonging to a particular category. These operations demonstrated how Pandas can efficiently retrieve meaningful subsets of data from large datasets.

The dataset was also checked for duplicate records using the duplicated() function. Duplicate entries can introduce bias and inaccuracies during analysis, so they were removed using the drop_duplicates() function. This improved the overall quality and reliability of the dataset.

To demonstrate feature engineering, a new derived column named **total_amount** was created. This column was calculated by multiplying the **price** and **quantity** values for each record. Since the Sample Superstore dataset does not directly contain a price column, the unit price was first derived by dividing Sales by Quantity, and then the total_amount column was generated. Creating derived features is an important step in data preprocessing because it allows additional useful information to be extracted from existing data.

Finally, the cleaned and processed dataset was saved as a new CSV file using the to_csv() function. Saving the cleaned dataset preserves all preprocessing changes and makes the data ready for future analysis, visualization, or machine learning tasks.

Overall, this assignment provided practical experience with the essential features of the Pandas library, including data loading, data exploration, missing value handling, filtering, duplicate removal, feature engineering, and exporting cleaned data. These are fundamental skills in data science, data analytics, and machine learning workflows, as clean and well-structured data forms the foundation for accurate analysis and reliable predictive models. Through this assignment, a strong understanding was developed of how real-world datasets are prepared before they are used for advanced analytical or machine learning applications.